

SCENTHUB

Project Proposal, Feasibility and Business Considerations

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1.Executive Summary

ScentHub is a software project being developed for a perfume business owned by **Yussif Kangre Abdulai**. At the moment, the business relies on manual methods to keep track of products, sales and customer information. While this has worked in the past, it has become harder to manage as the business continues to grow.

The aim of this project is to develop a simple and reliable system that will help the business manage its daily activities more efficiently. The system will allow the owner to keep accurate stock records, record sales, manage customer information and generate useful reports when needed.

This proposal explains the problem faced by the business, the solution that will be developed, the people who will use the system and the main requirements that will guide the project. It also outlines the project's scope, possible risks and the plan for completing the work throughout the semester.

2.Client and Business Background

ScentHub is a software project being developed for a perfume business owned by **Yussif Kangre Abdulai**. At the moment, the business relies on manual methods to keep track of products, sales and customer information. While this has worked in the past, it has become harder to manage as the business continues to grow.

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3.Problem Statement

Like many small businesses, ScentHub currently depends on manual record keeping for its daily operations. Sales, inventory updates and customer information are managed without a central system, making it difficult to keep records organised and up to date.

As the business grows, this way of working becomes less effective. Stock can be difficult to monitor, sales records may contain errors and finding past information can take longer than

necessary. These challenges can affect the efficiency of the business and make it harder for the owner to make informed decisions.

To address these challenges, the business needs a system that stores information in one place, keeps records accurate and makes everyday tasks quicker and easier to manage.

4. Proposed Solution

To solve the challenges faced by the business, our team will develop a web-based management system called **ScentHub**. The system will bring together important business activities such as inventory management, sales recording and customer record management into one platform.

Instead of relying on manual records, the business owner will be able to add new products, update stock levels, record sales and view important business information whenever it is needed. The system will also reduce repetitive work and make it easier to find and manage records.

Our goal is to create a system that is easy to use, reliable and practical for the client's daily operations. By doing this, we hope to improve efficiency, reduce errors and support better decision-making within the business.

5. Project Scope

In Scope:

- User login and authentication
- Inventory management
- Product management
- Sales recording
- Customer management
- Report generation
- Viewing stock availability
 - Updating product information

Out of Scope:

- Online payment processing
- Mobile application development
- Delivery management
- Integration with external payment systems
- Artificial intelligence recommendations
- Multi-branch business management

The project will focus on delivering the core features needed to manage the client's perfume business effectively within the semester. Features outside the agreed scope may be considered in future versions of the system but will not be included in this project.

6.Stakeholders and Users

Stakeholders	Role	Needs	Benefits
Business Owner	Owner of ScentHub	Manage inventory, sales and reports	Better control over business operations
Sales Attendant	Uses the system daily	Record sales and update stock	Faster and more accurate transactions
Customers	Purchase perfumes	Efficient service	Improved shopping experience
Development Team	Builds the system	Understand client requirements	Successfully deliver the project

7.Preliminary Functional Requirements

Functional requirements describe the tasks that the system must be able to perform. These requirements represent the main features that will help the client manage the business more effectively.

Requirement ID	Functional Requirement
FR1	The system shall allow users to log in securely using a username and password.
FR2	The system shall allow the business owner to add new perfume products.
FR3	The system shall allow users to update product information such as price, quantity and description.
FR4	The system shall allow users to delete products that are no longer available.
FR5	The system shall record every sale made in the business.
FR6	The system shall automatically update stock levels after each sale.

FR7	The system shall allow users to search for products quickly.
FR8	The system shall store and manage customer information.
FR9	The system shall generate reports showing sales and inventory information.
FR10	The system shall allow authorized users to view the system dashboard.

8. Preliminary Non-Functional Requirements

Non-functional requirements describe how well the system should perform rather than the features it provides.

Category	Requirement
Usability	The system should have a simple and user-friendly interface that is easy to learn.
Performance	Most pages should load within a few seconds during normal use.
Security	Only authorized users should be able to access the system through secure login credentials.
Reliability	Business records should be stored accurately without unnecessary data loss.
Availability	The system should be available whenever the business is operating, except during planned maintenance.
Maintainability	The system should be easy to update and improve as business needs change.

9. Main System Components

The ScentHub system will be divided into several components, with each one responsible for a specific part of the application's functionality.

Login Module

Handles user authentication and controls access to the system.

Inventory Management Module

Allows users to add, edit, delete and monitor perfume products and stock levels.

Sales Management Module

Records customer purchases and automatically updates inventory after every sale.

Customer Management Module

Stores customer information and makes it easy to retrieve customer records when needed.

Reporting Module

Generates reports on sales, inventory and other useful business information to support decision-making.

Database

Stores all information related to users, products, customers and sales in a secure and organized manner.

10. Project Milestones

The project will be developed using an incremental approach, with each phase building on the work completed in the previous stage.

Milestone	Planned Activities	Expected Outcome
D0	Team formation, repository setup and project planning	Project initialized and team roles assigned
D1	Project proposal and feasibility study	Proposal approved and project scope defined
D2	Requirements gathering and Software Requirements Specification (SRS)	Functional and non-functional requirements documented
D3	Product backlog, wireframes and Sprint 1 planning	Development plan and user interface prepared
D4	System architecture and CI/CD setup	Technical foundation established
D5	Sprint 1 implementation	First working prototype completed
D6	Sprint 2 implementation and testing	Additional features completed and quality improved
D7	Sprint 3 implementation, security and AI audit	System strengthened with security and monitoring
D8	Release candidate and user manual	Nearly completed system prepared for final review
D9	Final demonstration and client handover	Working system presented to the client
D10	Project closure and final documentation	Repository finalized and project officially completed

11. Business, Legal and Ethical Considerations

Developing ScentHub involves more than creating software. The team must also consider the responsibilities that come with managing business information.

Business Considerations

The system should improve the client's daily operations without making tasks more complicated. It should also be simple enough for users with limited technical experience to learn and use comfortably.

Legal Considerations

Customer and business information should be handled responsibly. Any personal information collected by the system must only be used for business purposes and should not be shared without permission. The team will also respect software licensing requirements when using third-party tools or libraries.

Ethical Considerations

The team will protect the confidentiality of the client's information throughout the project. Any use of artificial intelligence during development will follow the course guidelines, and all AI-assisted work will be reviewed and verified by team members before being included in the project.

12. Risk Analysis

The following table identifies some of the possible risks that may affect the project and the steps the team will take to reduce their impact.

Risk	Possible Impact	Mitigation Strategy
Client is unavailable for meetings	Requirements may become unclear	Arrange meetings early and communicate through email or WhatsApp when necessary
Team member becomes unavailable	Development work may slow down	Share responsibilities and keep documentation up to date
Technical challenges	Delays during implementation	Research solutions early and seek guidance when necessary

Data loss	Loss of important project work	Use GitHub regularly and maintain backup copies of documents
Misunderstood requirements	Features may not meet the client's needs	Confirm requirements with the client before implementation
Bugs during development	Reduced system quality	Test features regularly throughout development
Poor time management	Missed deadlines	Follow the project schedule and monitor progress weekly
Internet connectivity issues	Difficulty collaborating online	Complete work offline when possible and synchronize later
AI-generated errors	Incorrect code or documentation	Review, test and verify all AI-assisted work before using it
Scope changes	Increased workload and project delays	Discuss changes with the client and prioritize essential features

13. Conclusion

This proposal has outlined the purpose, scope and initial plan for developing the ScentHub system. The project is intended to provide a practical solution for improving inventory management, sales recording and customer management within the client's perfume business.

By following a structured software engineering process and working closely with the client throughout the semester, the team aims to deliver a reliable and user-friendly system that meets the agreed requirements. The next stage of the project will focus on gathering detailed requirements and producing the Software Requirements Specification (SRS), which will guide the design and implementation of the system.